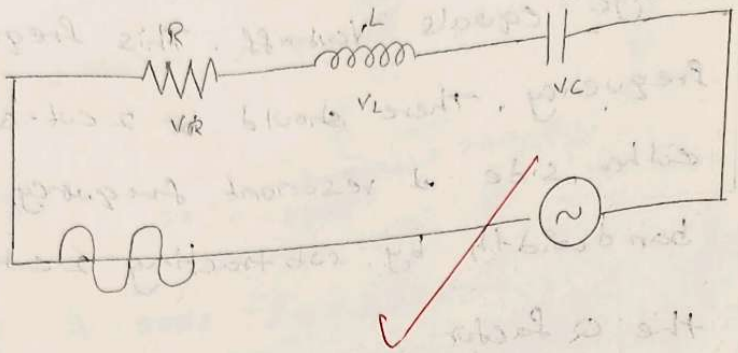
Circuit Diagram :-

Choose R = 10 Ω and C = 0.1 μF



Ex. No:
Date:

Transient Analysis of series RLC circuit

Aim:

To determine transient response of a series RLC circuit using Multisim simulator.

Apparatus Required:

S.No	compounds	specifications	quantity
1.	CRO	10 mHz	1
2.	DCB, DTB	---	1
3.	Resistors	10 KΩ	
4.	Bread board	---	1
5	connecting wires	---	---
6.	PC with multisim simulator	---	---

PROCEDURE:-

1. Make connection as per the circuit diagram.
2. Calculate the value of $v_c(t)$ and wire in table. $V_R(t)$ and $V_L(t)$

Compare the calculated values and measured values of voltage in both cases.

Tabulation :-

Voltage across capacitor :-

Vin Amplitude: _____ Vin Frequency: _____

Time constant $\tau = RC =$ _____

Time (t)	$V_C(t)$ - Charging cycle (V) observed value
$t=0$	0 V
$t=\tau$	2.5 V
$t=2\tau$	3.6 V
$t=3\tau$	4.3 V
$t=4\tau$	4.7 V
$t=5\tau$	5 V

Voltage across inductor :-

Vin Amplitude: _____ Vin frequency: _____

Time constant $\tau = L/R =$ _____

time (t)	$V_L(t)$ - Voltage inductor observed value
$t=0$	5 V
$t=\tau$	3.0 V
$t=2\tau$	1.8 V
$t=3\tau$	1.0 V
$t=4\tau$	0.5 V
$t=5\tau$	0 V

time (t)	$V_R(t)$ - Voltage resistor observed value
$t=0$	0 V
$t=\tau$	2 V
$t=2\tau$	3 V
$t=3\tau$	4 V
$t=4\tau$	4.5 V
$t=5\tau$	5 V

S.No	Description	Marks allotted	marks obtained
1	Experimental setup	20	20
2.	observation	20	20
3.	calculation	20	18
4.	viva	20	18
5.	Record	20	18
6.	Total	100	94

Result :

Thus the transient analysis of series RLC circuit is done practically and verified with theoretical value using multisim simulator.